



**University of
East London**

Pioneering Futures Since 1898

SCHOOL OF ARCHITECTURE, COMPUTING & ENGINEERING

Submission instructions

- All group member ID's to be included on the first page of assignment.
- All pages to be numbered sequentially
- All work has to be presented in a ready to submit state upon arrival at the ACE Helpdesk. Assignment cover sheets or stationery will **NOT** be provided by Helpdesk staff

Module code	CN4000		
Module title	Information Systems Modelling and Design		
Module leader	Arish Siddiqui		
Assignment tutor	Arish Siddiqui, Solomon Alexis, Dhara Parekh, Mahmud , Hisham Abougrad, Madhav, Mayank, Galvin, Reena, Jasmi, Abdul Qadoos, Pedram, Manasa.		
Assignment title	Coursework: Group Assignment		
Assignment number	One		
Weighting	80%		
Handout date	20-OCT-2025		
Submission date	19-DEC-2025		
Learning outcomes assessed by this assignment	1-8 in the Module Spec		
Turnitin submission requirement	Yes	Turnitin Grade Mark feedback used?	No
UEL Plus Grade Book submission used?	No	UEL Plus Grade Book feedback used?	
Other electronic system used?		Are submissions/feedback totally electronic?	Yes
Additional information			



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Form of assessment:

☐ Individual work ☒ Group work

For **group work** assessment which requires members to submit both individual and group work aspects for the assignment, the work should be submitted as:

☒ Consolidated single document ☐ Separately by each member

Number of assignment copies required:

☒ 3 ☐ 2 Other ☐ Other

Assignment to be presented in the following format:

- ☒ On-line submission
- ☐ Stapled once in the top left-hand corner
- ☐ Glue bound
- ☐ Spiral bound
- ☐ Placed in a A4 ring bound folder (not lever arch)

Note: To students submitting work on A3/A2 boards, work has to be contained in suitable protective case to ensure any damage to work is avoided.

Soft copy:

- ☐ (to be attached to the work in an envelope or purpose made wallet adhered to the rear) adhered to the rear)
- ☐ wallet adhered to the rear) adhered to the rear)
- ☒ Soft copy not required

Note to all students

Plagiarism is a serious offence --please do not jeopardise your degree and please submit your assignment before the due date.

Group Coursework– 80%

Please read the whole assignment to ensure you understand all the requirements before you begin.

This assessment should be attempted in groups of **4 students**.

You must keep regular minutes of any meetings that you have while undertaking this course work as they will be required if there is a dispute about any members' contribution. Your minutes should:

- be dated
- include a list of all group members present and apologies for any absences
- include allocation of work to be achieved by each member by the next meeting
- include the time, date and venue of the next meeting
- be signed to indicate the agreement of all group members, present or not.

All individuals will receive the group grade for this coursework unless clear evidence is provided of unequal contributions.

The marking criterion for the Coursework is based on the assessment criteria at the end.

Deliverables

Assignment Report of no more than 2500 words (80 %)

Coursework Brief: Smart Travel and Holiday Booking Agency

Deliverables

Scenario: GlobeTrek Smart Travel Agency

You have been appointed as a Technology Consultant for *GlobeTrek Travel Agency*, a rapidly growing company that specialises in personalised travel experiences.

GlobeTrek operates both online and through physical branches across the UK and Europe. It offers customers a full range of travel services including flight bookings, hotel reservations, local tours and experiences, and travel insurance packages.

The company's current systems are outdated, with several manual processes still in use for booking management, customer communication, and payment tracking.

Management is therefore seeking to develop an integrated smart travel management system that improves efficiency, enhances the customer experience, and ensures data security across all platforms.

Your role is to analyse the existing systems, recommend improvements, and design a new integrated information system suitable for *GlobeTrek's* operational needs.

Task 1: Understanding the Organisation and Its Information Systems

Task 1.a (5 Marks)

Briefly introduce *GlobeTrek Travel Agency* — its structure, operations, and services.

Identify the types of Information Systems (IS) and IT specialists that the organisation currently uses or would likely need to support its operations — both in-house (e.g., IT administrators, data analysts, system developers) and outsourced (e.g., web hosting, cloud services, payment gateway providers).

Select one of the systems currently in use — for example, *Online Booking System*, *Customer Relationship Management (CRM)*, or *Inventory and Supplier System*.

Describe the system in detail, identifying its hardware, software, data, and people resources.

Task 1.b (5 Marks)

Select one of the systems identified in Task 1.a (e.g., the *Online Booking and Payment Platform*).

Perform a system analysis and suggest improvements that could make the system more efficient and effective in achieving business goals such as faster bookings, secure payments, or improved customer satisfaction.

Task 2: System Development and Design Methodologies (10 Marks)

Based on your findings, propose a new system to replace or enhance the existing one, for example, a *Smart Integrated Travel Management System (SITMS)* that connects flights, hotels, and activities on one platform.

Explain the System Development Life Cycle (SDLC) and the information system modelling/design methodology (e.g., *Agile*, *Scrum*, or *Waterfall*) you would apply in developing the system.

Justify your choice of methodology based on *GlobeTrek's* needs, team structure, and customer interaction requirements.

Task 3: Data Security, Ethics, and Legal Compliance (5 Marks)

GlobeTrek manages large volumes of customer data including personal details, passport information, and payment records.

Discuss the data security measures that should be implemented to safeguard this sensitive information (e.g., encryption, firewalls, access control, data backups).

Explain the social, ethical, and legal issues involved in managing such data, and how the organisation must comply with UK GDPR and Data Protection Act 2018 to ensure data privacy and customer trust.

Task 4: Project Planning and Team Collaboration (5 Marks)

Demonstrate how you would plan and manage the system development project.

- Task 4.a – MS Project: Explain how MS Project can be used to track and monitor the project timeline, resources, and dependencies.
- Task 4.b – Work Breakdown Structure (WBS) and Gantt Chart: Create or outline these tools to show how the project tasks are structured and scheduled.
- Task 4.c – Minutes of Meetings: Describe the purpose of meeting minutes in maintaining accountability and progress tracking.
- Task 4.d – Task Allocation: Create a table showing how project tasks would be distributed among team members.
- Task 4.e – Task Evaluation: Reflect briefly on how each group member evaluated their own contribution and teamwork.

Task 5: System Modelling and Design

Your proposed system should meet *GlobeTrek's* operational needs.

The company may have multiple branches across different countries, each with managers, customer service agents, and travel consultants who assist clients both in person and online. The system must handle flight bookings, hotel reservations, tours and activities, payments, cancellations, and customer feedback. It should also cater to special customer types such as VIP clients and corporate travellers.

Task 5.a – Entity Relationship Diagram (ERD) (10 Marks)

Identify and illustrate the entities required for the proposed system (e.g., *Customer, Flight, Hotel, Tour, Payment, Branch*).

Show the relationships between entities using an ERD (you may use *Draw.io, Lucidchart, or MS Access*).

Include any assumptions you make.

Task 5.b – Schema and Attributes (5 Marks)

List all attributes for each entity and present them in a schema table.

Clearly identify:

- Primary Keys (PK) and Foreign Keys (FK)
- Data Types (e.g., INT, VARCHAR, DATE)
- Null/Not Null constraints

Task 5.c – Use Cases and Data Flow Diagrams (10 Marks)

Conduct a requirements analysis to define how users interact with the system.

Create Use Case Diagrams and case descriptions for processes such as:

- Booking a flight or hotel
- Cancelling or modifying reservations
- Making payments
- Viewing invoices or itineraries

Also include a Level 0 DFD, and optionally a Level 1 DFD for a specific process (e.g., *Process Booking or Manage Payments*).

Consider security and access privileges, for example:

- Customers can view bookings and download itineraries.
- Staff can update bookings and process payments.
- Managers can authorise refunds and access reports.

Task 5.d – Academic Integrity and Presentation (5 Marks)

Your work must demonstrate academic writing standards, appropriate Harvard Referencing, and clear evidence of academic integrity (e.g., Turnitin originality report).

Task 6. Presentation and Presentation slides (20 Marks)

Please include your presentation slides into the report.

Short Report elements – Assessment Criteria	Grade Band
Excellent coverage of issues with good exemplification and a complete list of references along with Turnitin report.	70-100% Excellent
Good coverage of issues with some relevant exemplification with adequate references.	60-69% Good to Very Good
Adequate coverage of issues with little exemplification	50-59% Satisfactory to Good
Unclear coverage of issues but little exemplification	40-49% Pass standard
No relevant material or very little relevant material	0-39 Fail

Deliverable – ERD	Grade Band
There is a clear indication in the answer that students have fully understood the problem domain and spent a considerable amount of time iterating over different solutions. All relevant entities, data attribute, and relationships have been identified and many-to-many relationships have been eliminated. Documentation is to a professional standard along with a complete list of references.	70-100% Excellent
Most of the relevant entities, data attributes and relationships have been identified. Many-to-many relationships have been eliminated. Any assumptions have been justified in the context of the case study and problem domain. Documentation is of a high standard and there is an evidence of iteration and adequate referencing.	60-69% Good to Very Good
About half of the relevant entities, data attributes and relationships have been identified. The standard of documentation is satisfactory. Evidence of iteration will be patchy.	50-59% Satisfactory to Good
Very little or no iteration. Less than half of the relevant entities, data attributes and relationships have been identified. The standard of documentation is generally weak.	40-49% Pass standard to satisfactory
Significant errors and misunderstandings. Entities, data attributes, and relationships have not been identified correctly. Level of detail is inappropriate. Documentation is poor. Task not attempted or incomplete or fails to identify the most obvious points.	0-39 Fail
Deliverable – Use Case	Grade Band

There is a clear indication in the answer that students have fully understood the problem domain and spent a considerable amount of time iterating over different solutions. All relevant actors and use cases have been identified and their relationships clearly shown. Documentation is to a professional standard.	70-100% Excellent
Most of the relevant actors and use cases have been identified. Any assumptions will be justified in the context of the case study and problem domain. Documentation will be of a high standard and there will be an evidence of iteration.	60-69% Good to Very Good
About half of the actors and use cases have been identified and there will be no redundant use cases. The standard of documentation is satisfactory. Evidence of iteration will be patchy.	50-59% Satisfactory to Good
Very little or no iteration. There may be a number of redundant use cases. The standard of documentation is generally weak.	40-49% Pass standard to satisfactory
Significant errors and misunderstandings. Actors and use cases have not been identified correctly. Level of detail is inappropriate. Documentation is poor. Task not attempted or incomplete or fails to identify the most obvious points.	0-39 Fail

Assessment form for Presentation:

UNIVERSITY OF EAST LONDON

Group:

School of Computing, Architecture and Engineering

CN4000 Information Systems Modelling and Design (20%)

Students to fill this information and email it to tutor before the presentation: Examiners will not be liable for any mistakes in student ids.

Group No:

Group Member (Student No):

**Do all students agree to equal distribution of marks? : Yes/No IF
No state percentage for each.**

The following is a checklist of things that **MUST** be present in order that this assessment point may go ahead. If any of the items are missing the student is deemed to have **FAILED**. It is the group's responsibility to have brought the items on the checklist with them.

Item	Present	Not Present
ALL Group Members		
Necessary resources		
All students agreed on the distribution of marks		

Assessment

Please mark the following criteria. All marks are out of a total of 20.

Clear, concise and all the group members played an active part		3 Marks
Business introduction with WBS		4 Marks
Project plan with Gantt Char		2 Marks
Use Case		4 Marks
ERD		4 Marks
Sufficient explanations and descriptions		3 Marks

Overall Mark:

Assessors should show their comments on the back of this form.

Signed:

As Assessors

Date: